Application of machine learning techniques to prediction of stock market indicies

To what extent can machine learning predict the daily movements of the KOSPI composite index?

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This extended essay evaluates the use of a Neural Network in the one-day prediction of the KOSPI composite index—stock index for the Korea Exchange (KRX). Here, a novel method of combining past values of the KOSPI index and microeconomic and macroeconomic factors in prediction is introduced.

First, raw data for the KOSPI index and eight microeconomic and macroeconomic factors from 2004 to 2022—including past prices of the KOSPI index and the S&P 500 index—believed to be correlated with the KOSPI index were collected. The data were detrended using log first difference. The Granger Causality Test was then used to identify a selection of three factors most correlated with the KOSPI index. A deep neural network involving dense and Long-Short-term memory (LSTM) layers was then created using the Keras library. The neural network was trained against the collected data to predict whether the KOSPI index would exhibit a bull (index increase) or bear (index decrease) move the next day.

Results suggest that the model was able to predict the future direction of the KOSPI index's movement with a success rate of 62.0%. This figure outperforms any of the naïve methods, including random walk (50.0%), always predicting bear (46.8%) and always predicting bull (53.5%). Further discussions on the results suggest that it may be difficult to significantly increase the accuracy of the model, considering the efficient market hypothesis and random walk hypothesis.